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Access the code, data, and analysis at https://github.com/jossl95/co-pub_networks

Intersectional Inequalities in Co-Authorship Networks: Gender, Origin, and Collaboration in Dutch Social Science

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1 Introduction

Human knowledge is created, sustained, and spread in an inherently social way. Departmental colleagues, disciplinary conferences, publishing venues and editors, and varieties of academic committees are all central to academic norms and practices (Bourdieu, 1988; Burt, 1998; Stark et al., 2020). Yet simultaneously, academic socialization also constitutes a central pathway through which inequalities emerge and reproduce. For instance, access to prestigious academic positions depend heavily on one’s mentors (Hofstra et al., 2022); dissertation committees exhibit gender-homophilous tendencies in awarding cum laude graduate degrees (Bol, 2023); and the size and composition of co-authorship teams dictate whether papers are disruptive or consolidating in their findings (Wu et al., 2019). Here we argue that paper co-authorships in science are perhaps a key social organization in academe that contributes to and sustains academic inequalities. This is because (a) (peer-reviewed) papers are the essential form to contemporary human knowledge production that (b) is increasingly done in co-authorship teams (REFS) that (c) dynamically shapes productivity and visibility (Abbasi et al., 2011; Ductor, 2015; Li et al., 2022; Moody, 2004; Newman, 2001; Wallace et al., 2012). As such, co-authorship and the resulting output is still the main aspect on which scientists are valued and less access to (more successful) co-authors causes long-lasting negative effects on academic careers.

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Along what axes do co-authorships cause disparities? Particularly problematic ones are divides by gender and ethnic origin: when women, minoritized scholars, or minoritized women scholars find less access to prolific co-authorship, it is one more signal of unequal access to academic careers and it offers narrower epistemological perspectives on our understanding of what good scientific knowledge is. It also signifies reproduced advantage instead of distributed opportunity, and this counters an ideal of a scientific meritocracy (Merton, 1973). We say “one more,” because gender disparities in academic careers are well-documented on many dimensions already (e.g., Barrios et al., 2013; Bol, 2023; Greska, 2023; L. Kim et al., 2022; Lutter & Schröder, 2016): women have lower salaries (C. Kim & Shin, 2024), obtain fewer grants (Stadmark et al., 2020), receive less recognition for often more innovative work (Hofstra et al., 2020), have shorter and less prolific careers (Mulders et al., 2024; Tolsma et al., 2025), and work on topics that yield lower career returns (C. Kim & Shin, 2024). Additionally, minoritized scholars – by national origin, ethnic background, and skin color are less likely to obtain senior positions, have fewer publications and citations, and face slower career progression (Hofstra et al., 2020; Kozlowski et al., 2022; Whittington, 2018).

How do (co-authorship) networks and gender and national origin interrelate in academia? Prior work shows that women are less likely than men to occupy powerful positions in (co-authorship) networks (Gao et al., 2022; Whittington, 2018; Whittington et al., 2024). Women’s collaborations are also more often local and not international (Kwiek & Roszka, 2021), and they populate fewer senior authorship positions (Barrios et al., 2013). Furthermore, scholars with minoritized nationalities to occupy peripheral network positions (Hâncean et al., 2014; Khor & Yu, 2016; J. Kim et al., 2016; Kwiek & Roszka, 2021), limiting exposure to high-impact journals and influential collaborators. Origin-based disadvantages are reinforced by institutional prestige hierarchies (Allison & Long, 1987, 1990; Wapman et al., 2022): work from elite universities and English-speaking or Western countries is more likely to be indexed, cited, and recognised (Khor & Yu, 2016; Li et al., 2022), while research from less visible institutions or non-core regions is underrepresented (Céspedes et al., 2025; Zheng et al., 2025).

Here, we build on earlier insights and examine to what extent network mechanisms – such as homophily, triadic closure, preferential attachment, and brokerage – shape collaboration opportunities and inequalities (Barabasi & Albert, 1999; Bojanowski & Czerniawska, 2020; Ebadi & Schiffauerova, 2015; Gondal, 2011, 2018) and reproduce gender and origin-based segregation in collaboration networks (Gao et al., 2022; Hâncean et al., 2014; Kwiek & Roszka, 2021; Whittington et al., 2024). Therefore, we ask: *to what extent are there gender- and origin-based differences in collaboration (networks)?* and *to what extent are there gender- and origin-based differences in homophily?* *Do actors tend to choose collaborators who share their gender or origin?* We do so by drawing from comprehensive data on a near-population of social scientists in Dutch universities and their co-authorship histories. These time-stamped snapshots function as an anchorpoint in each scholar’s career, fixing their affiliation and co-authorships at a given moment in time and

thereby reducing the influence of survivor and publication bias that typically distort conventional bibliometric network studies (Simko, 2015). This allows us to characterise disparities in collaboration opportunities and outcomes not just for a selected discipline or journal, but for the entire social-science habitat, mirroring recent full-population approaches in the science-of-science literature (Hofstra et al., 2020; Whittington et al., 2024). Prior work interprets gender or origin gaps in collaboration without fully considering network endogenous processes such as transitivity (Gao et al., 2022; Kwiek & Roszka, 2021), possibly overestimating key indicators like homophily. Here, we are among the first to explicitly model path dependency and endogenous network processes such as transitivity by using longitudinal network models. This shifts the focus from descriptive differences to the network processes through which inequalities are reproduced. Because our data provide rich information on all actors in the network and approximate a full-population sample, we can also address survivor bias and sample selection problems that arise in bibliometric studies focused only on highly productive scholars or selected journals (Hâncean et al., 2014; Ossenblok et al., 2014).

We also posit that gender- and origin-based differences in collaboration do not happen in a vacuum and introduce the idea of intersectionality in academic collaboration networks. This provides a framework for understanding how axes of inequality intersect, emphasizing that gender and origin (or race, ethnicity, nationality) are mutually constitutive axes of social stratification (Crenshaw, 1989; King, 1988). This is important, because prior work in science and higher education for instance demonstrates that gender penalties tend to be more pronounced for women from underrepresented nationalities or ethnic groups, while shared nationality may mitigate disadvantages for majority-group women – or vice versa (Kwiek & Roszka, 2021; van der Wal et al., 2021). Approaches that simply consider gender and origin therefore understate the disadvantage that accrues at their intersection and risk missing the specific penalties experienced by intersectional groups (Kozłowski et al., 2022). Therefore, we are among the firsts to explicitly contribute an intersectional perspective to study collaboration opportunities and segregation in collaboration networks among Dutch social scientists. This enables us to ask and answer novel questions, for instance on whether homophily on gender and origin is stronger for majority-group men than for minoritised scholars, whether international women are particularly excluded from triadic closure processes, and whether bridging ties to high-status alters are unequally distributed (Kwiek & Roszka, 2021; Whittington et al., 2024). By focusing on these mechanisms by means of our intersectional perspective, we identify in far more detail than before where in the collaboration process intersectional penalties emerge. Therefore, we ask: *to what extent does gender- and origin-based homophily intersect? Do actors tend to choose collaborators who share both their gender and their origin?*

Finally, we examine the role of social connections in academic networks even further by explicitly studying second order social capital influences. This refers to the resources and opportunities people can access through the contacts of their contacts. This type of capital may be particularly relevant in academic networks because introductions to influential collaborators, invitations to special issues,

or access to well-funded research teams may arise from the networks of mentors, co-authors, or departmental colleagues. Women and minoritized scholars may be less likely to benefit from the indirect networks of their advisors or collaborators. As a result, inequalities compound across network levels, making second-order social capital a key mechanism through which gender, origin, and intersectional disparities in collaboration emerge. To our knowledge, such questions have hardly been asked in the context of co-authorship networks despite reasonable assumptions that such an approach offers us far more insight into how unequal access to high-status, well-connected, and highly productive alters-of-alter shapes cumulative advantage and disadvantage in academic careers. We therefore study to what extent network aspects of second-order social capital - in particular the h-index, degree centrality, and academic position - operate differently across gender and origin groups and their intersection?

2 Data and Methods

2.1 Data

This study introduces a database of scholars and their publications that approximates the full population of staff employed in sociology and political science departments at Dutch research universities in recent years. We reconstructed a roster of department employees at three time points - 19 December 2022, 19 April 2024, and 12 October 2025 - by manually collecting staff lists from public departmental webpages. For each snapshot, individual and departmental information was stored in harmonized Excel files with separate sheets for sociology and political science. This data set contains for every listed staff member, their name, university, job title, and email address, for each of the three measurement waves.

To link these scholars to their publication histories, we use OpenAlex as a bibliometric backbone (Priem et al., 2022). OpenAlex is a free, openly accessible scholarly metadata source that offers coverage and reference statistics broadly comparable to major proprietary databases such as Web of Science and Scopus, with particularly strong coverage of persistent identifiers (e.g. ORCID), while also providing a more balanced linguistic coverage than Web of Science, albeit with some remaining limitations in the accuracy and completeness of language metadata (see, e.g., Céspedes et al., 2025; Culbert et al., 2025). First, we constructed a set of institution records for all universities in our sample by assembling, for each institution, its preferred English and native names and using these variants to query the OpenAlex API for the corresponding institution entities. This procedure yields disambiguated institution identifiers, which we then attach to all occurrences of these universities in the scholar roster. Second, OpenAlex author profiles are retrieved for each scholar by searching on their harmonized name and, where possible, constraining the search to the known institution identifiers. Our matching procedure explicitly allows each scholar to be linked to multiple OpenAlex author IDs, which is important because some individuals maintain more than

one profile in OpenAlex due to ambiguous or shifting name forms across their publications. Third, the resulting candidate author profiles are filtered using a combination of institutional overlap, name similarity thresholds, and a small set of manually curated exclusion rules. For every candidate author, the script computes a Jaro–Winkler similarity score between the OpenAlex display name and the query name and unpacks information on last known institutions, country codes, and persistent author identifiers. The implicit aim of this step is to systematically exclude false matches and retain a high-confidence mapping between our scholars and OpenAlex author IDs. Fourth, all works associated with the selected author IDs are downloaded from OpenAlex and transformed into a tidy format in which each row represents a unique author-work combination, with compact author and work identifiers, publication year, document type, and coauthor information. Together, the scholar roster and linked OpenAlex records form the basis for the co-publishing networks analyzed in this paper. all code and intermediate outputs are documented and made available through the public replication package website.

i Note

Add note on sample selection and sample size. Potentially add flow-chart

2.2 Variables

2.2.1 Dependent Variable

Collaboration, the dependent variable, was operationalized as a binary co-publishing tie between pairs of scholars. Using the OpenAlex-based publication data described above, we identified all works in which any two scholars in our roster appeared together in the authorship list. For each pair of scholars, we then coded a tie as present (1) if they had coauthored at least one publication within the observation window, and absent (0) otherwise. This information was assembled into a series of adjacency matrices in which nodes represent scholars and undirected edges represent observed co-publishing relationships. These longitudinal collaboration networks form the outcome in our models, capturing both the existence and evolution of co-authorship ties over time.

2.2.2 Independent Variables

Gender was operationalized as an inferred gender indicator based on scholars' first names. We first used the **genderizeR** package to query the genderize.io database, which provides probabilistic gender assignments for first names based on large-scale social media data (Wais, 2016, see also genderize.io documentation). The underlying database aggregates first-name and gender information for more than one hundred million individuals across a wide range of countries, and returns both a predicted gender and associated probability for each name. Using this API, we obtained an initial classification for all unique first names in our scholar roster. To improve coverage for names that are common in the Netherlands and reduce misclassification, we then matched first names to the Dutch public civil registry name database hosted by the Meertens Instituut (Nederlandse

Voornamenbank), which contains all officially registered first names in the Netherlands and their distribution by sex over time (Meertens Instituut, n.d.). Where available, this registry information was used to validate or refine the genderize.io-based assignments, especially for names with low counts or ambiguous probabilities in genderize.io.

Origin was operationalized as an inferred country-of-origin indicator based on scholars' surnames. We first used the **DemografixeR** package to query the nationalize.io API, which provides probabilistic nationality assignments for last names based on large-scale web data (Brenninkmeijer, 2025, see also nationalize.io documentation). The underlying database aggregates name-country occurrences from diverse online sources, and returns, for each surname, a set of country codes with associated probabilities. Using this API, we obtained an initial nationality distribution for all unique surnames in our scholar roster. To improve coverage for surnames that are common in the Netherlands and to correct systematic misclassifications, we then complemented these predictions with information from the Dutch Database of Surnames in the Netherlands (Nederlandse Familienamenbank), which reports the number of name bearers per surname and provides etymological notes on its origin and meaning (Centrum voor familiegeschiedenis, n.d.). A custom scraping and parsing pipeline extracted origin-related signals from these etymology texts (e.g., language adjectives such as "Nederlandse", "Turkse", or "Marokkaanse", references to specific countries or regions, and indicators of Dutch surname particles), and combined them with national occurrence counts. A manually curated patch file was used to correct ambiguous or frequently misclassified surnames. In a final harmonization step, we collapsed the various signals into a single cleaned origin label (e.g., "Nederlandse", "Turkse", "Marokkaanse", "Chinese") and mapped these labels to ISO 3166-1 alpha-2 country codes. The resulting variable captures, for each scholar, the most likely country-of-origin signal implied by their surname.

i Note

Add note over aggregation of this variable

Position was operationalized as an indicator of scholars' academic rank/role within the Dutch university system. Starting from the free-text job titles collected from departmental staff pages, we first cleaned and standardized these titles (e.g., "universitair docent", "universitair hoofddocent", "hoogleraar", "postdoc-onderzoeker", "promovendus") and translated them into a common English terminology. We then mapped all titles to a set of seven analytically meaningful categories: *PhD student*, *Postdoc*, *Lecturer*, *Researcher*, *Assistant professor*, *Associate professor*, and *Full professor*. In cases where multiple roles were listed for the same individual (e.g., combined teaching and research appointments or endowed/visiting positions), we applied a simple priority hierarchy that privileges structural career rank (professor > associate professor > assistant professor > researcher/postdoc

> lecturer > PhD student). The resulting harmonized variable therefore captures each scholar's primary academic position.

h-index was used as a bibliometric indicator of scholars' cumulative research impact. For each scholar, we drew on the OpenAlex publication and citation data linked to their author identifiers and calculated the Hirsch index, defined as the largest integer h such that the scholar has at least h publications with $\geq h$ citations each (Hirsch, 2005). We first assembled, for every scholar, the full set of works retrieved from OpenAlex, along with their citation counts at the time of data collection. These works were then sorted in descending order of citations, and the h-index was computed by identifying the point at which the number of publications equals or exceeds the citation threshold. The resulting measure reflects each scholar's career-wide citation impact within the coverage of OpenAlex over the observation window and is used as a continuous indicator of scholarly standing in our analyses.

Note

Currently the H-index is calculated for the full sample, but could be calculated to vary with time. What has your preference?

2.2.3 Control Variables

Affiliation captures scholars' primary institutional location and is measured as a categorical indicator of the Dutch research university at which they are employed. Based on the initial manual scrape of departmental staff pages, we distinguish eight universities: Radboud University, Erasmus University Rotterdam, University of Amsterdam, Leiden University, Vrije Universiteit Amsterdam, Utrecht University, Tilburg University, and University of Groningen. We include affiliation as a control variable to account for structural differences between universities in size, funding, and collaboration culture that may shape co-publishing opportunities.

Discipline indicates whether a scholar is employed in sociology or political science, derived from the manually coded departmental information in the initial staff rosters and coded as a binary indicator for sociology versus political science. We include discipline as a control variable to capture field-specific norms in publishing, collaboration rates, and typical team sizes that could otherwise confound observed differences in network positions and outcomes.

2.3 Analytical Approach

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